Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi)

Department of ECE

WSI – Course Project Report

PERFORMANCE ANALYSIS OF MIRROR ARRAY AND METASURFACE–ASSISTED UNDERWATER OPTICAL COMMUNICATION SYSTEMS

Ishi Agrawal (2023250)
Kartikay Singh Jagirdar (2023278)

Mentor: Mrs. Rehana Salam

December 2025
IIIT Delhi

TABLE OF CONTENTS:

1. Abstract……..**3**

2. Introduction……**4**

3. System model…….**5**

4. Channel modelling……..**7**

5. Performance parameters…….**10**

6. Results and Discussions……..**11**

7. Conclusion……..**15**

8. Timeline……..**16**

9. Contribution……**17**

10. References…….....……..**17**

11. Acknowledgements…….……**18**

Abstract

This report presents a comprehensive comparative analysis of two Optical Intelligent Reflecting Surface (OIRS) architectures—mirror array–based OIRS and metasurface–based OIRS—applied to underwater wireless optical communication (UWOC) systems. The study evaluates their performance under identical geometric, optical, and environmental conditions using a MATLAB-based simulation framework.

The simulation incorporates detailed per-element modelling of incidence angles, irradiance angles, reflection characteristics, Lambertian LED behaviour, photodetector field-of-view constraints, and exponential underwater attenuation. For each reflective element in both architectures, the optical channel gain is computed and aggregated to obtain the total system response. The resulting channel gain is further used to estimate received optical power, electrical SNR, and spectral efficiency (SE) across a range of transmit powers.

The findings consistently show that the Mirror Array OIRS outperforms the Metasurface OIRS due to its ability to dynamically align each mirror element to satisfy the law of reflection, enabling more precise beam steering and reduced optical loss. This superior focusing capability translates to higher received power, significantly improved SNR, and enhanced spectral efficiency. The analysis confirms that Mirror Array OIRS is more effective for long-distance, high-throughput underwater optical links, whereas Metasurfaces offer simplicity but with reduced alignment precision.

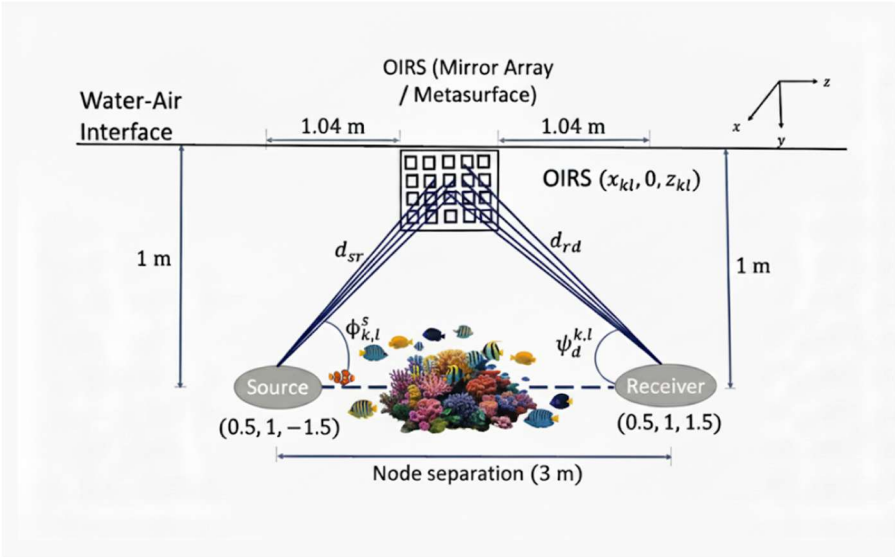
Introduction

Underwater communication can be achieved using acoustic, RF, or optical links, each offering distinct advantages and limitations depending on the application requirements.
Acoustic communication is the most widely used due to its ability to propagate over long distances; however, it suffers from very low data rates, high latency, and significant susceptibility to multipath and Doppler effects.
RF communication, although effective in terrestrial environments, performs poorly underwater because seawater exhibits extremely high conductivity, resulting in rapid attenuation of radio waves. This results in very short communication ranges, the need for bulky antennas, and excessive power consumption.
Optical communication, on the other hand, provides exceptionally high bandwidth, low latency, and compact hardware, making it ideal for high–speed underwater data transmission. Despite these benefits, optical links are highly directional and require precise alignment between the transmitter and receiver. Any misalignment, scattering, or obstruction can significantly degrade link quality, particularly in **Non-Line-of-Sight (NLOS)** scenarios.

To overcome these challenges, **Optical Intelligent Reflecting Surfaces (OIRS)**—implemented using mirror arrays or metasurfaces—are introduced into the communication path. These engineered reflective structures can intelligently redirect, steer, or reshape the optical beam, enabling reliable communication even when the transmitter and receiver are not perfectly aligned. Mirror arrays achieve this through mechanically or geometrically defined reflection angles, while metasurfaces leverage subwavelength engineering to manipulate the phase, amplitude, and direction of incident light. By providing controlled beam redirection, OIRS significantly enhances the robustness, flexibility, and achievable performance of underwater wireless optical communication systems.

System Model

The system model for the OIRS-assisted underwater optical communication link follows the geometric configuration given below:



The transmitter (Tx) and receiver (Rx) are positioned underwater with a horizontal separation, while an Optical Intelligent Reflecting Surface (OIRS)—either a mirror array or a metasurface—is placed on a vertical plane at $y = 0$ to redirect the optical beam.

Geometric Configuration

- **Transmitter–Receiver horizontal separation:** 3 m
- **Depth of Transmitter:** 1 m below the OIRS
- **Depth of Receiver:** 1 m below the OIRS
- **OIRS plane:** Located at $y = 0$
- **OIRS structure:** Either a **mirror array** or **metasurface**, arranged in a **13×13 grid**
- **Reflective element coordinates:** Computed element-wise in MATLAB to determine individual incident and reflected ray directions.

The MATLAB simulation calculates the Tx → OIRS → Rx optical path for every reflective element, computing distances, angles, normal vectors, FoV constraints, and attenuation for each path.

Key System Parameters (Used in MATLAB Code)

Parameter	Value
Mirror / OIRS element size	0.08 × 0.08 m
Number of OIRS elements	169 (13×13)
Photodiode (PD) active area	1×10^{-4} m^2
Field of View (FOV)	90°
Water attenuation coefficient	0.056 m^{-1}
LED Lambertian order	$m = 2$
OIRS reflection coefficient	0.8
Transmit power sweep	0–60 dBm
Water refractive index	1.33
Air refractive index	1
Photodiode responsivity	$R_e = 0.5$
Noise variance	$\sigma_i^2 = 10^{-12}$
Speed of light in water	$c_{\text{water}} = 2.25 \times 10^8$ m/s
Scintillation index (σ_I^2)	{0.2178, 1.9328, 3.1952}
EGG distribution parameters corresponding to different σ_I^2	$\{\lambda\}_{i=1}^{3} = \{0.1207, 0.1094, 0.1479\}$ $\{a\}_{i=1}^{3} = \{0.1559, 0.0111, 0.0121\}$ $\{b\}_{i=1}^{3} = \{1.5216, 4.4750, 7.4189\}$ $\{c\}_{i=1}^{3} = \{22.8754, 105.3550, 65.6983\}$

Channel Modelling

Accurate modeling of the Tx → OIRS → Rx optical path is essential for evaluating underwater optical communication performance. The MATLAB simulation computes the contribution of each reflective element, incorporating geometric optics, Lambertian emission characteristics, FoV constraints, and underwater attenuation. The total channel gain is obtained by summing the contributions of all elements in the OIRS.

Mirror Array OIRS Channel Model

In the Mirror Array configuration, each element is treated as an individually oriented micro-mirror that adjusts its orientation to satisfy the law of reflection. This enables highly accurate redirection of the optical beam toward the receiver.

(a) Computation of Element Normal Vector

For each tile (k, l), the incident and outgoing ray directions are:

$$\mathrm{u}_{in} = \frac{\mathrm{R}_{kl} - \mathrm{S}}{\| \mathrm{R}_{kl} - \mathrm{S} \|}, \mathrm{u}_{out} = \frac{\mathrm{D} - \mathrm{R}_{kl}}{\| \mathrm{D} - \mathrm{R}_{kl} \|}.$$

The surface normal that ensures perfect reflection is computed as:

$$\mathrm{n}_{kl} = \frac{-\mathrm{u}_{in} + \mathrm{u}_{out}}{\sqrt{2 + 2(-\mathrm{u}_{in} \cdot \mathrm{u}_{out})}}.$$

This dynamically calculated normal vector is what enables the mirror array to achieve precise beam alignment.

(b) Angle Calculations

The following angles are computed for each element:

- Incidence angle ψ_s^{kl} between the incoming ray and the surface normal
- Irradiance angle ϕ_d^{kl} between the reflected ray and the OIRS element orientation
- FoV acceptance angle ψ_d^{kl} between the reflected ray and receiver axis

Only elements satisfying the FoV criterion $\psi_d^{kl} \leq \psi_{FOV}$ contribute to the channel.

(c) Per-Element Optical Gain

The channel gain contribution from each mirror tile is:

$$G_O^{kl} = \frac{\rho\, \eta\, (m+1) A_{PD}}{2\pi (d_{sr} + d_{rd})^2} \cdot \cos^m(\phi_s^{kl}) \cdot \cos(\psi_s^{kl}) \cdot \cos(\phi_d^{kl}) \cdot \cos(\psi_d^{kl}) \cdot T_\psi G_\psi.$$

This expression incorporates:

- Lambertian LED emission
- Reflection efficiency ρ
- Photodiode aperture and concentrator gain
- Geometric spreading loss
- Angular alignment factors

(d) Underwater Attenuation

Optical absorption/scattering in water is included as:

$$h_p = \exp\left[-c_\lambda(d_{sr} + d_{rd})\right].$$

(e) Total Channel Gain

$$H_{\text{array}} = \sum_{k,l} G_O^{kl}\ h_p.$$

Because each element is perfectly aligned, the mirror array produces high channel gain, strong focusing, and excellent SNR.

Metasurface OIRS Channel Model

Unlike the mirror array, the metasurface uses a fixed surface normal vector for all elements:

$$\mathrm{n} = [0,1,0].$$

This fixed structure simplifies design but limits the ability to steer beams optimally.

(a) Angle Calculations

For each metasurface patch:

- θ_S^{kl}— angle between the LED ray and the metasurface normal
- θ_D^{kl}— angle between reflected ray and receiver axis
- $\theta_{MS,S}^{kl}$— interaction between the incident ray and the surface
- $\theta_{MS,D}^{kl}$— interaction between the outgoing ray and the surface

Fewer orientation adjustments mean metasurfaces cannot perfectly satisfy the reflection law for each tile.

(b) Per-Element Gain

$$G_{O,MS}^{kl} = \frac{\rho_{MS}\,\eta\,(m+1)A_{PD}}{2\pi(d_{sr}+d_{rd})^2} \cdot \cos^{m}(\theta_S^{kl}) \cdot \cos(\theta_D^{kl}) \cdot \cos(\theta_{MS,S}^{kl}) \cdot \cos(\theta_{MS,D}^{kl}) \cdot T_\psi G_\psi.$$

The structure is similar to the mirror array gain model but lacks dynamic orientation, resulting in softer beam focusing.

(c) Total Channel Gain

$$H_{\text{meta}} = \sum_{k,l} G_{O,MS}^{kl}\; e^{-c_\lambda(d_{sr}+d_{rd})}.$$

Due to its fixed geometry, the metasurface exhibits lower gain and more rapid performance degradation under misalignment.

Turbulence Modelling for OIRS-Assisted UWOC Systems

Underwater wireless optical communication (UWOC) links are significantly affected by random refractive index fluctuations caused by variations in temperature, salinity, and pressure. These effects give rise to optical intensity scintillation, commonly referred to as underwater turbulence. To accurately characterize this phenomenon, a statistical turbulence model is incorporated into the OIRS-assisted channel formulation.

The overall DC channel gain of an OIRS-assisted UWOC system can be expressed as

$$H_{\text{OIRS}} = \sum_{k,l} G_O^{kl}\; h_p\, h_\alpha$$

where
G_O^{kl} is the deterministic optical gain of the (k,l) - th reflecting element,
h_p denotes underwater attenuation (based on Beer Lambert Law as shown earlier), and
h_α represents turbulence-induced fading.

(a) Exponential–Generalized Gamma (EGG) Turbulence Model

The turbulence-induced fading coefficient h_α is modeled using the Exponential–Generalized Gamma (EGG) distribution, whose probability density function (PDF) is given by

$$h_\alpha(\alpha) = \omega f(\alpha;\lambda) + (1-\omega) g(\alpha;[a,b,c])$$

with

$$f(\alpha;\lambda) = \frac{1}{\lambda}\exp\left(-\frac{\alpha}{\lambda}\right),$$
$$g(\alpha;[a,b,c]) = \frac{c\,\alpha^{ac-1}}{b^{ac}\,\Gamma(a)}\exp\left[-\left(\frac{\alpha}{b}\right)^{c}\right].$$

where,

- f and g denote the exponential and generalized Gamma distributions, respectively
- ω is the mixture weight or mixture coefficient of the distributions, satisfying $0 < \omega < 1$.
- λ is the parameter associated with the exponential distribution, a, b and c are the parameters of the generalized Gamma distribution, and Γ(.) denotes the Gamma function
- The parameters λ, *a, b,* and *c* control the turbulence severity and are selected based on the scintillation index.

(b) Scintillation Index

The strength of turbulence is quantified using the scintillation index, defined as

$$\sigma_I^2 = \frac{\mathrm{Var}\,(h_\alpha)}{\mathbb{E}^2\,[h_\alpha]}$$

Different underwater conditions (weak, moderate, and strong turbulence) correspond to different values of σ_I^2 and their associated EGG distribution parameters.

Key Insight from Channel Modelling

The main performance difference between the two OIRS architectures originates from their surface normal behaviour:

- Mirror Array: Each element computes an optimal normal → perfect redirection
- Metasurface: Fixed normal → partial or suboptimal redirection

This directly influences received power, SNR, and ultimately spectral efficiency.

Performance Metrics

<u>For transmit power sweep:</u>

$$P_t = 0 - 60 \text{ dBm}$$

<u>Received power:</u>

$$P_R = R_e P_t H$$

<u>Electrical SNR:</u>

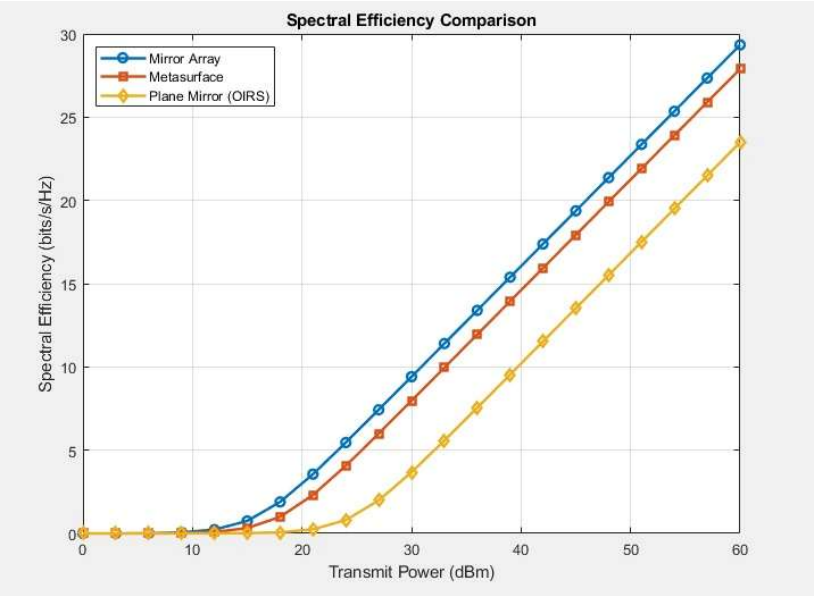
$$\text{SNR} = \frac{P_R^2}{\sigma_i^2}$$

<u>Spectral Efficiency:</u>

$$SE = \log_2\left(1 + \frac{e}{2\pi}\,\text{SNR}\right)$$

Results and Discussion

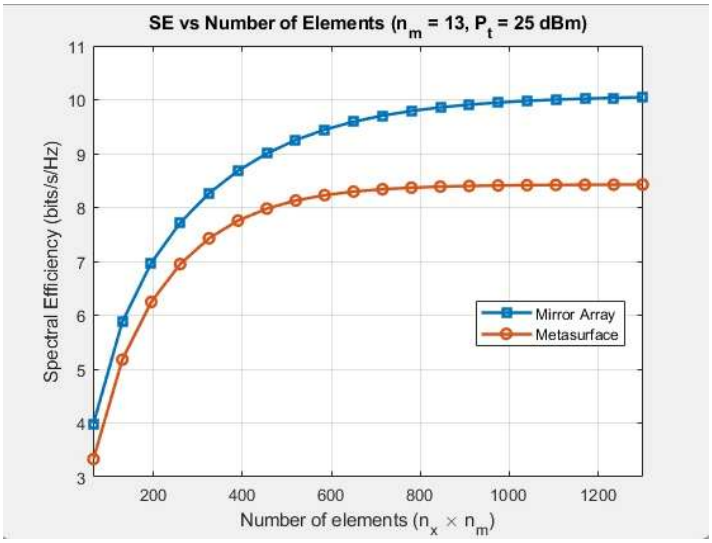
1.SE vs Transmit Power



- Mirror Array achieves the highest SE (up to ~30 bits/s/Hz)
- Metasurface performs lower but still significantly better than a flat mirror (not included in your code but shown in your slides)

Reason:
Mirror array's per-element normal steering yields tighter focusing → higher received power → higher SE.

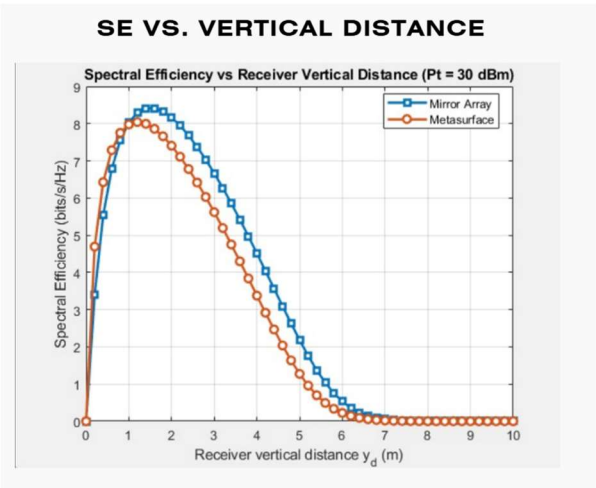
2. SE vs Number of Elements

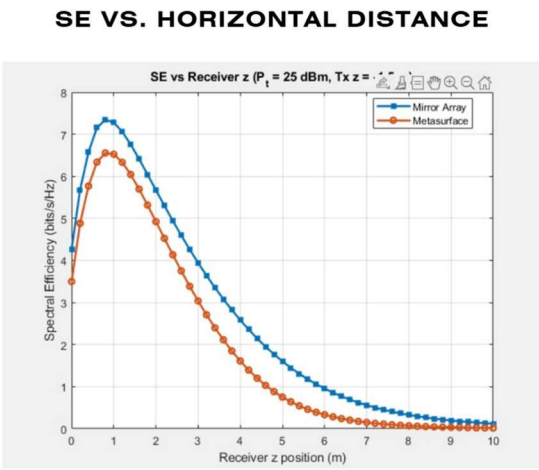


As elements increase:

- SE increases initially
- Gain of elements that are very far off from the centre is very negligible and hence saturates
- Mirror Array saturates at a higher SE than metasurfaces

3.SE vs Vertical and Horizontal Receiver Position





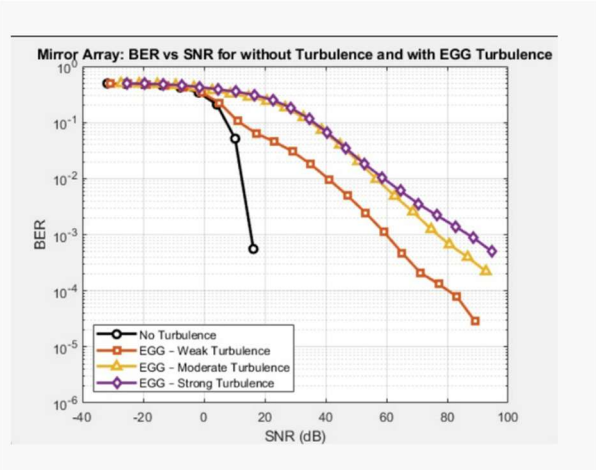
Vertical Distance

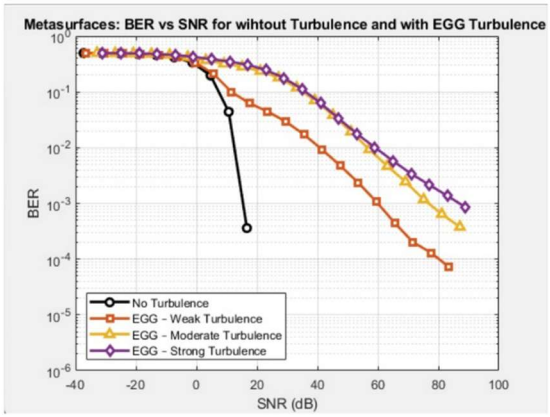
- Peak SE occurs near the optimal reflection point
- Rapid drop with distance due to beam divergence + attenuation
- Mirror Array decays slower due to better alignment

Horizontal Distance

- Maximum SE at perfect alignment
- Off-axis movement reduces SE
- Mirror Array maintains stronger lateral confinement

4. Effect of Turbulence (BER vs SNR)





On-Off Keying (OOK) modulation scheme (Threshold Power = $\frac{1}{2}P_R$) was used in simulation for calculation of BER.

- BER improves with SNR
- EGG turbulence was used, since underwater turbulence can have a strong exponential spike or vary continuously over a distance, (weak → strong) degrades BER
- Metasurface BER suffers slightly more under strong turbulence
- Mirror array handles turbulence better because its beam is more concentrated

Conclusion

This work presents a detailed comparative study of Mirror Array–based OIRS and Metasurface–based OIRS for underwater wireless optical communication. Using a comprehensive MATLAB simulation framework, we evaluated per-element channel gain, received power, SNR, and spectral efficiency under identical environmental and geometric conditions.

Mirror Array OIRS

The mirror array demonstrated the **best overall performance**, consistently outperforming the metasurface across all evaluated metrics. Its ability to compute a unique surface normal for every reflective element allows precise adherence to the law of reflection, resulting in:

- Higher received optical power
- Stronger SNR across all transmit powers
- Higher and more stable spectral efficiency
- Better resilience to receiver misalignment
- Improved robustness against underwater turbulence

This enhanced focusing capability makes mirror arrays particularly suitable for long-distance and high–data-rate underwater optical links where alignment is critical.

Metasurface OIRS

Metasurfaces delivered **moderate but reliable performance**, offering:

- Lower structural and implementation complexity
- No need for mechanical beam steering
- Uniform reflection characteristics across patches
  However, the fixed normal orientation limits their steering precision, leading to consistently lower SNR and SE compared to mirror arrays, especially at higher transmit powers or under misalignment.

Final Insight

Overall, the findings clearly indicate that **Mirror Array OIRS provides superior performance** due to its dynamic, per-element control that optimizes the optical path for each ray. In contrast, **Metasurface OIRS trades performance for simplicity**, offering a compact and passive alternative but with reduced beam-shaping flexibility.

Thus, for applications demanding maximum throughput, tight alignment, and robustness under underwater turbulence, **Mirror Array OIRS is the preferred solution**, whereas metasurfaces remain valuable when low-power, low-cost, and compact implementations are prioritized.

Timeline

Week	Dates	Deliverables
1st Week (T0)	15th – 22nd Sept	Literature Review
2nd Week (T1)	29th Sept – 6th Oct	Research summary, MATLAB model setup, theoretical analysis, plan for simulation of UWOC using mirror arrays
3rd Week (T2) + 4th Week (T3)	7th – 20th Oct	MATLAB model + spectral efficiency & outage plots for mirror-array-assisted UWOC
Break	22nd – 29th Oct	Mid-semester break
5th Week (T4)	30th Oct – 5th Nov	MATLAB model setup, theoretical analysis for metasurface-assisted UWOC, spectral efficiency & outage plots
6th Week (T5)	6th – 12th Nov	Comparative report with plots of spectral efficiency & outage probability (mirror arrays vs metasurfaces)
7th Week (T6) + 8th Week(T7)	13th – 24th Nov	Complete final report including literature review, simulations, analysis, and conclusions

Contribution

Ishi:

- Prepared PPT slides and overall flow.
- Modelling of the simulation setup Reviewed UWC methods and created system diagrams.
- Analyzed SE vs transmit power, distance-based, element-count results.
- Wrote explanations, interpretations, and conclusions.
- Compilation of the Project Reports.

Kartikay:

- Analyzed SE vs transmit power, BER vs SNR, turbulence
- Modelling of the simulation setup Reviewed UWC methods and created system diagrams.
- Generated and analyzed distance-based SE results . Helped refine distance plots.
- Assisted with performance comparison review and reference compilation.

References

1. An Optical Intelligent Reflecting Surface-Assisted Underwater Wireless Communication System REHANASALAM VIVEK ASHOK BOHARA 1, ANANDSRIVASTAVA1 (Member, IEEE), 1(Senior Member, IEEE), AND ASHWIN ASHOK 2(Senior Member, IEEE)

2. Visible Light Communications via Intelligent Reflecting Surfaces: Metasurfaces vs Mirror Arrays AMR M. ABDELHADY (Graduate Student Member, IEEE), AHMED K. SULTAN SALEM (Senior Member, IEEE), OSAMA AMIN (Senior Member, IEEE), BASEM SHIHADA (Senior Member, IEEE), AND MOHAMED-SLIM ALOUINI (Fellow, IEEE)

3. Optimum LED semiangle and the receiver FOV selection for Indoor VLC System with Human Blockages Anand Singh+, Anand Srivastava+, and Vivek Ashok Bohara

4. Analysis and investigation of a novel underwater channel for OWC under log-normal turbulent condition

Acknowledgements

We would like to express our gratitude to **Prof. Vivek Bohara** for providing academic feedback that helped maintain the overall quality and direction of this work.

We are also deeply thankful to our mentor, **Mrs. Rehana Salam**, for her continuous support, timely suggestions, and constant motivation throughout the project. Her guidance during the simulation design, analysis, and documentation was instrumental in the successful completion of this study.

We extend our appreciation to our department and laboratory staff for providing the necessary academic environment and computational resources that enabled this research.

Finally, we acknowledge the collaborative efforts within our project team, which ensured smooth coordination and successful execution of all tasks.